

Examples of linear and non-linear models

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1 Linear models excluding State-Space models

List of SCM-model parameters:

- $\mathbf{w}$ the linear parameters of the model,
- $\mathbf{v}$ the non-linear parameters of the model,
- the covariance (mutual information) of the variables $X_1, \dots, X_n$,
- 1. the covariance of the linear part of the parameters $\mathbf{w}$,
- 2. the covariance of the non-linear part of the parameters $\mathbf{v}$.

The transformations

- 1. only G to f it admissible (there is no conversion from the model to G)

Architectures of the linear models:

- 1. linear model $\hat{\mathbf{x}} = \mathbf{W}^T \mathbf{x}$: a single matrix (2-index) that maps $X_1, \dots, X_n$ to $\hat{X}_1, \dots, \hat{X}_n$,
- 2. generalized linear model $\sigma \circ \mathbf{W}^T \mathbf{x}$: non-parametric link function of the linear model,
- 3. special non-linear model $g_1(\mathbf{v}_i) \circ g_r(\mathbf{v}_r)(\mathbf{x})$: a model with functions $g_i(\mathbf{v}_i)(X_j)$,
- 4. general non-linear model $\mathbf{W}^T g_i(\mathbf{v}_i)(X_j)$ (obsoleted): a linear combination of the model with functions,
- 5. generalized non-linear model $\sigma \circ \mathbf{W}^T g_i(\mathbf{v}_i)(X_j)$: a non-linear non-parametric transformation of the linear combination

Here the notation $\circ$ means the composition of functions. It could be replaced by brackets.